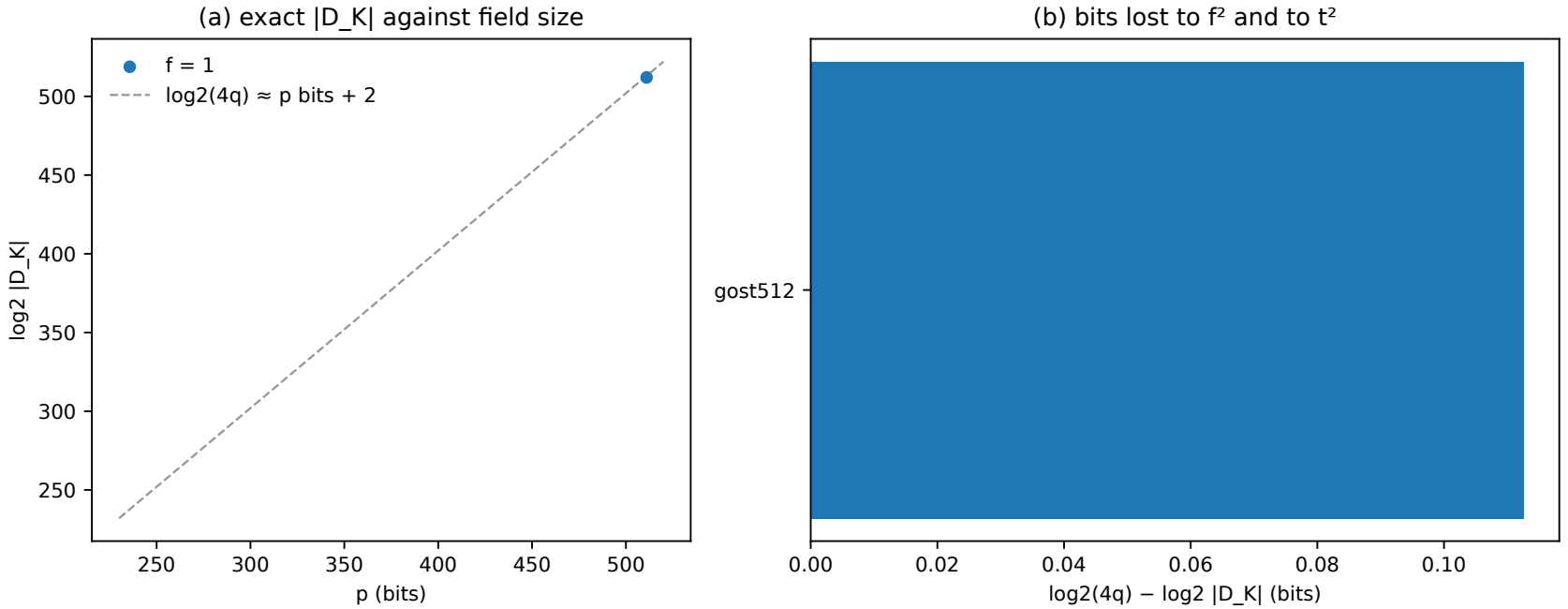
Exact CM discriminant of gost512 (batch 5), and the four composites still open

research/endosweep_dk_exact_batch5_20261023 (2026-10-23). Data: `dk_exact.json`, `factorizations.jsonl`, `open_composites.json`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration.

Part A. gost/gost512 is the one curve completed between the fourth frozen driver snapshot (batch 4) and the fifth (2026-10-23). Its $4q - t^2$ (155 digits) is $19870558523 \cdot p_{71} \cdot p_{74}$. The 11-digit factor was found by GMP-ECM at B1 = 11000. **The split of the 144-digit cofactor into $p_{71} \cdot p_{74}$ was not found by this project:** it was fetched from FactorDB (number id 1100000009030548498, fetched 2026-10-22T23:04:32Z), after the driver's ECM had run the complete B1 = 3·10^6 tier (2400 curves) and 1200 of 4480 curves at B1 = 11·10^6 on it without a factor. FactorDB does not say who found the split. Both primes were re-proved here and the product re-checked against $4q - t^2$ recomputed from the curve parameters. std-curves records no `cm_disc`, so D_K was not checked against the literature.



n = 1 curve; exact integers (PARI isprime-proved factorisations), no uncertainty.

Figure 1: (a) $\log_2|D_K|$ against p for gost512 ($f = 1$). (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`.

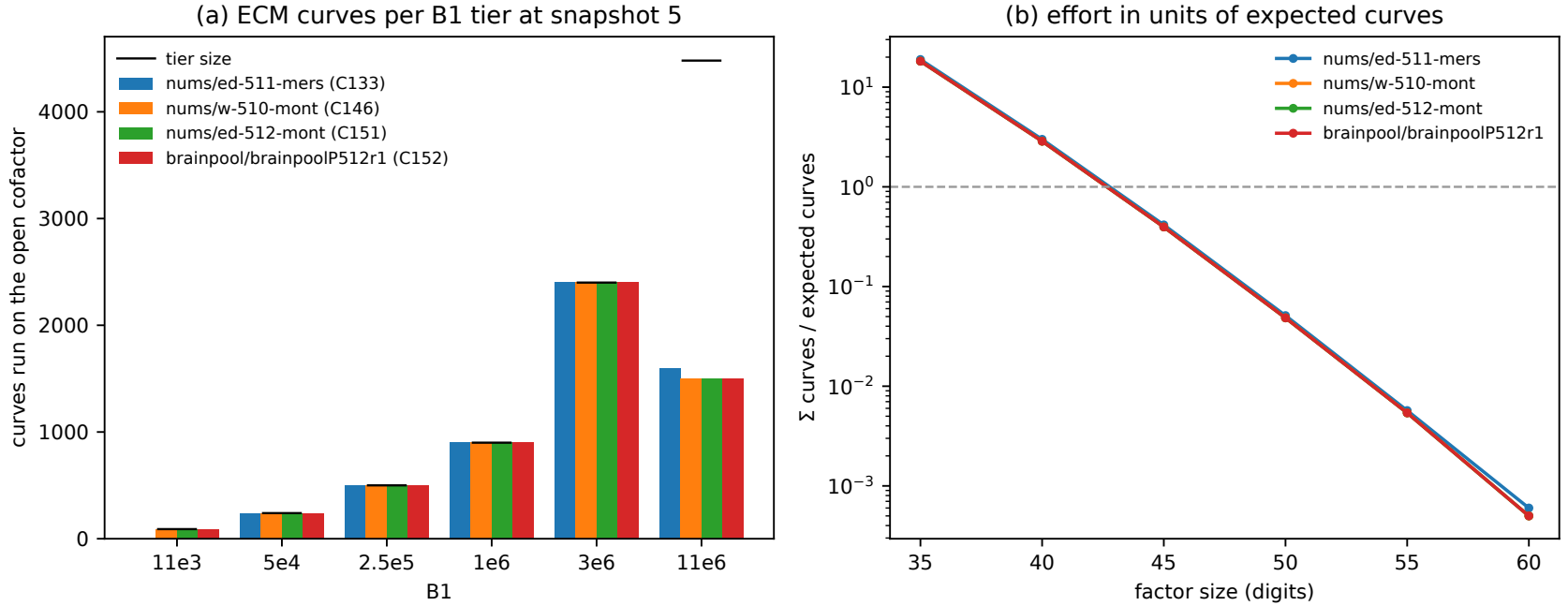
curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
gost/gost512	511	155	155	1	(155 digits)	512	2^510	no cm_disc

Lean: `CryptoResearch.Endosweep.DkExactBatch5.exact_discriminants` (target `TASK-20261023-6fca5a`), reusing `DkExact.DkCert`; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).

Part B: the four $4q - t^2$ still open at snapshot 5

No D_K is stated for these. Curves per B1 are those the driver state records; the coverage columns divide them by GMP-ECM 7.0.5's own expected-curve table (`ecm -v`) and give $\exp(-\sum)$, the approximate chance that a factor of that size, if one exists, was missed (an idealised model). ed-511-mers' C133 has been on CADO-NFS since 2026-10-22T23:05Z (in progress); the C146 to C152 are beyond practical NFS on this 4-core machine.

curve	cofactor	11e3	5e4	2.5e5	1e6	3e6	11e6 (of 4480)	Σ/E 40 dig.	Σ/E 45 dig.	FactorDB
nums/ed-511-mers	C133	≤ 90 (n.r.)	240	500	900	2400	1600	3	0.42	C
nums/w-510-mont	C146	90	240	500	900	2400	1500	2.87	0.4	C
nums/ed-512-mont	C151	90	240	500	900	2400	1500	2.87	0.4	C
brainpool/brainpoolP512r1	C152	90	240	500	900	2400	1500	2.87	0.4	C



Counts: driver_state.snapshot5.json. Expected curves: GMP-ECM 7.0.5 ecm -v. Dashed: 1 expected find. ed-511-mers' 11e3 curves not recorded (drawn as 0).

Figure 2: (a) ECM curves per B1 tier on each open cofactor at snapshot 5. (b) The same effort in units of GMP-ECM's expected number of curves per factor size; 1 (dashed) is one expected find.